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Software Design Patterns
section 02
Lab 3-Part 2

Part 2: Game Level

The Source Code

```
interface GameLevel extends Cloneable {
    GameLevel clone();
    void setName(String n);
    String getName();
}

class EasyLevel implements GameLevel {
    private String name, terrain;

    public EasyLevel(String name, String terrain) {
        this.name = name; this.terrain = terrain;
    }

    private EasyLevel(EasyLevel other) {
        this(other.name, other.terrain);
    }

    @Override public EasyLevel clone() { return new EasyLevel(this); }
    @Override public void setName(String n) { this.name = n; }
    @Override public String getName() { return name; }

    @Override public String toString() { return "EasyLevel{name=" + name + ",
terrain=" + terrain + "}"; }
}

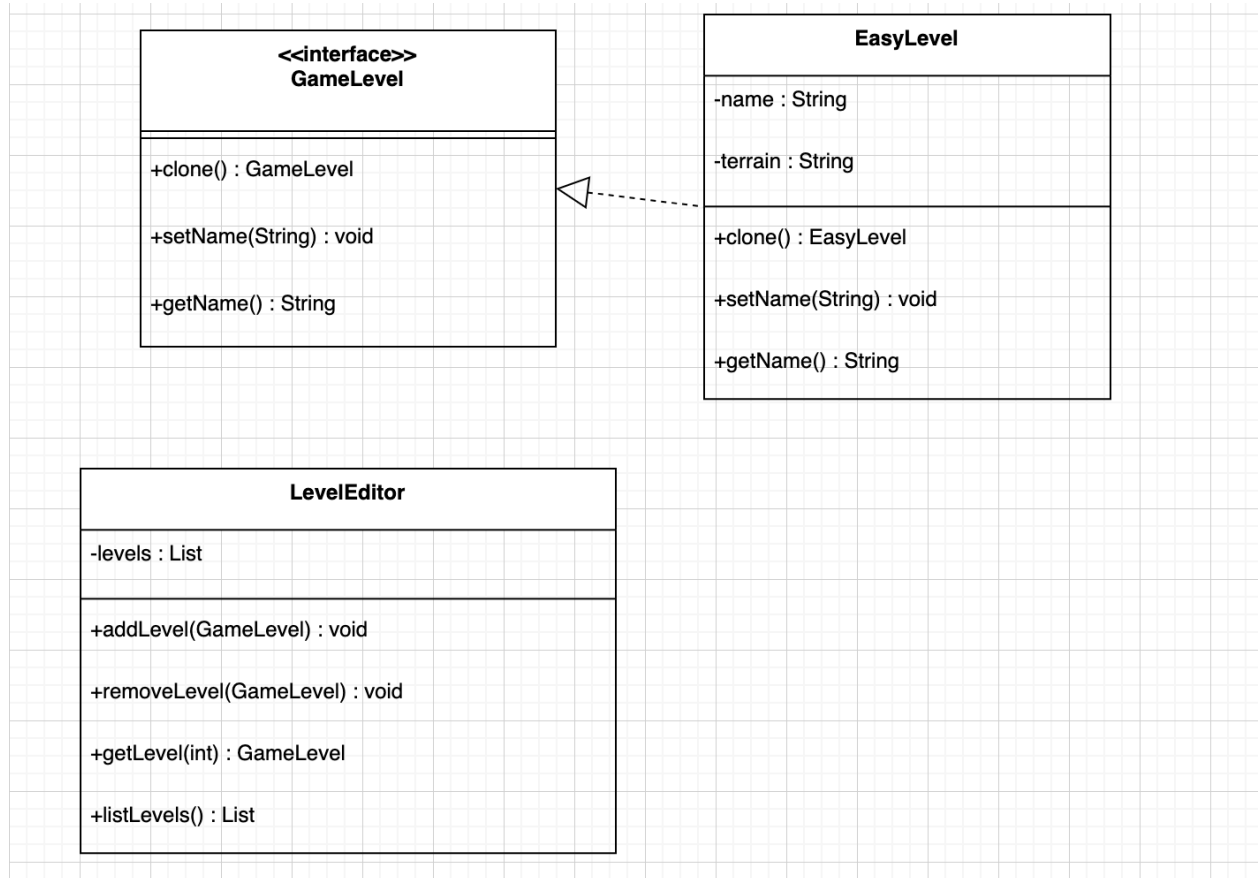
public class LevelEditor {
    public static void main(String[] args) {
        EasyLevel base = new EasyLevel("Tutorial Meadow", "forest");

        EasyLevel l1 = base.clone();
        l1.setName("Desert Adventure");

        EasyLevel l2 = base.clone();
        l2.setName("Underwater World");

        System.out.println(l1);
        System.out.println(l2);
    }
}
```

The Class Diagram



The Output

```
EasyLevel{name=Desert Adventure, terrain=forest}
EasyLevel{name=Underwater World, terrain=forest}
```